Max Fung

Senior Product Engineer

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Experience

Founding Engineer · Mynatek (via Studio Makai) — May 2025 – Sep 2026

- Sole product engineer at a HealthTech startup building a computer vision–based facility management platform for hospitals, from MVP through a $1.8M ARR pilot deal
- Owned the customer-facing layer: a React dashboard integrating real-time CV pipeline data, role-based access, and operational workflows; built the Go device management and notification services and moved the stack to Connect-RPC
- Built mobile PWA workflows for hospital staff: QR-based session management, real-time HLS video from Raspberry Pi edge devices, and a templated workflow pattern enabling rapid iteration on cleaner and technician user flows
- Drove product direction alongside CTO and CEO, translating high-fidelity design specs into iterative, shippable scope under tight pilot deadlines

Lead Product Engineer · Dirac — Apr 2024 – Jan 2025

- Drove a ground-up architecture overhaul after identifying a fundamentally broken data model, rallying the company around a full rebuild; led a 4-engineer frontend team through it and shipped the v1 3D manufacturing interface, foundational work that enabled Dirac's $4M multi-year Anduril partnership
- Architected Dirac's core document model, reconciling hierarchical CAD B-Rep/Assembly trees into a linear, temporal instruction sequence; built a custom tree library with per-operation dirty-node tracking and Redux middleware that synced changes through a Redis write-back cache into DynamoDB, with native undo/redo
- Hand-rolled a virtualized tree component rendering 500k+ CAD nodes with desktop-native UX; built the core UI library and gRPC integration with Go backends
- Drove product direction, architecture, and visual design end to end in the absence of dedicated product or UX teams, bridging engineering and business strategy

Software Engineer · Flexe — Feb 2022 – Nov 2023

- Self-taught Kotlin and claimed ownership of the Android WMS app as the only mobile developer on a 10-person senior-heavy team; rebuilt Pick and Sort to a new UX standard with components adopted across teams
- Migrated the picking service from a Rails monolith to Kotlin microservices, endpoint by endpoint

Integrated Product Team Leader · Boeing — Nov 2020 – Feb 2022

- While leading an integrated product team, built an unsanctioned data aggregation tool that became department-critical: reverse-engineering internal databases to integrate EPM, MES, and procurement data into automated daily reports, saving every PM 30–45 minutes daily
- Navigated internal resistance from the site's data analytics org to get the tool formally deployed on IT infrastructure, serving the entire El Segundo manufacturing site

Projects

Atlas · atlasemc.com

- Built a historical atlas of the EarthMC Minecraft server, any town or nation on any day, with no database: a Go collector archives the public API every 15 minutes and an exporter compiles it into static files on Cloudflare's edge
- Designed the export format for fast first paint (a small index, time-bucketed files, history sharded by ID, content-hashed names) and built the SolidJS frontend with a Rust/WebAssembly territory worker

Drumhaus · drumha.us

- Solo-built browser-native drum machine using Tone.js and the Web Audio API; features custom DSP, sub-sample scheduling, and a from-scratch animation system
- Engineered bit-packed preset compression, PWA architecture, and a responsive sequencer interface; shipped as an open creative tool with organic traction

Education

B.S., Mechanical Engineering · Loyola Marymount University — 2016 – 2020